2-CARTESIANNESS OF THE DIAGONAL SQUARE UNDER BASE CHANGE OF 2-FIBRE PRODUCTS

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Editorial note. Counted span is the author's own proof environment, source lines 6183-6204, following the statement at 6162-6181.

Editorial note. The proof is complete; every step of the chain is justified by a cited lemma or the defining 2-fibre product square.

Allowed standard prerequisites (editorial).

- Associativity of 2-fibre products. lemma-associativity-2-fibre-product, cited at proof line 6202. Supplies the middle reassociation step of the equivalence chain at proof lines 6187-6200.
- Erasing a factor in iterated 2-fibre products. lemma-2-fibre-product-erase-factor, cited at proof lines 6202-6203. Supplies the collapse X x_(X x_Y X) (X x_Y X x_Y V) = X x_Y V and the analogous factor erasures in the chain.

Source reference. Original lemma 02XD: sources/stacks-categories/pinned-original-source.tex, source lines 6052–6052. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02XC: sources/stacks-categories/pinned-original-source.tex, source lines 5982–5982. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let*

$$\begin{array}{ccc} \mathcal{U} & \longrightarrow & \mathcal{V} \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{Y} \end{array}$$

be a 2-fibre product of categories. Then the diagram

$$\begin{array}{ccc} \mathcal{U} & \longrightarrow & \mathcal{U} \times_{\mathcal{V}} \mathcal{U} \\ \downarrow & & \downarrow \\ \mathcal{X} & \longrightarrow & \mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \end{array}$$

is 2-cartesian.

Date: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This is a purely 2-category theoretic statement, valid in any $(2,1)$-category with 2-fibre products. Explicitly, it follows from the following chain of equivalences:

$$\begin{aligned}\mathcal{X} \times_{(\mathcal{X} \times_{\mathcal{Y}} \mathcal{X})} (\mathcal{U} \times_{\mathcal{V}} \mathcal{U}) &= \mathcal{X} \times_{(\mathcal{X} \times_{\mathcal{Y}} \mathcal{X})} ((\mathcal{X} \times_{\mathcal{Y}} \mathcal{V}) \times_{\mathcal{V}} (\mathcal{X} \times_{\mathcal{Y}} \mathcal{V})) \\ &= \mathcal{X} \times_{(\mathcal{X} \times_{\mathcal{Y}} \mathcal{X})} (\mathcal{X} \times_{\mathcal{Y}} \mathcal{X} \times_{\mathcal{Y}} \mathcal{V}) \\ &= \mathcal{X} \times_{\mathcal{Y}} \mathcal{V} = \mathcal{U}\end{aligned}$$

see Lemmas 02XC and 02XD. □